

AI & ML Task 3 Report: Model Validation, Overfitting Analysis and Hyperparameter Tuning

This report presents the validation and optimization of a Decision Tree Regressor used for house price prediction. The objective was to evaluate model performance, identify overfitting, apply cross-validation, perform hyperparameter tuning using GridSearchCV, and justify the selection of the final model.

1. Baseline Model Performance

A Decision Tree Regressor was initially trained using default parameters. The model was evaluated using RMSE (Root Mean Squared Error) and R^2 Score. The baseline model achieved an RMSE of 0.7030 and an R^2 score of 0.6228. While these results indicated reasonable predictive capability, additional validation was required to determine whether the model generalized well to unseen data.

2. Overfitting Analysis

The training score of the baseline model was 1.0000, while the testing score was 0.6228. This large gap indicates a classic case of overfitting. A training score of 1.0000 means the model learned the training data almost perfectly, including noise and dataset-specific patterns. However, its lower performance on testing data demonstrates reduced generalization ability. Overfitting occurs when a model becomes excessively complex and memorizes the training data instead of learning meaningful relationships. In the context of a Decision Tree Regressor, overfitting often occurs when the tree grows without sufficient restrictions. Deep trees can create highly specific decision rules that fit the training set exceptionally well but fail to perform consistently on new observations. Therefore, reducing model complexity becomes necessary to improve robustness and deployment readiness.

3. Cross-Validation Strategy

To obtain a more reliable estimate of model performance, 5-fold cross-validation was performed. The dataset was divided into five subsets, and the model was trained and validated multiple times using different folds. This approach reduces dependence on a single train-test split and provides a broader understanding of how the model behaves across different portions of the data. Cross-validation is important because a single split may accidentally produce overly optimistic or pessimistic results. By averaging performance across multiple folds, a more stable estimate of model quality is obtained.

4. Hyperparameter Tuning Approach

To address overfitting and improve predictive performance, GridSearchCV was used to perform systematic hyperparameter tuning. The following Decision Tree parameters were explored: • max_depth • min_samples_split • min_samples_leaf GridSearchCV evaluated multiple parameter combinations using cross-validation and selected the configuration that achieved the best validation performance. This method provides a structured and objective way to search for improved model configurations. The optimal hyperparameters identified were: max_depth = 10 min_samples_leaf = 4 min_samples_split = 2 These settings restrict excessive tree growth and help the model focus on meaningful patterns rather than memorizing training examples.

5. Tuned Model Performance

After applying the optimal hyperparameters, the tuned Decision Tree Regressor was evaluated again. Results: Baseline RMSE: 0.7030 Tuned RMSE: 0.6389 Baseline R² Score: 0.6228 Tuned R² Score: 0.6885 The decrease in RMSE indicates that prediction errors were reduced. The increase in R² score shows that the model explains a greater proportion of variance in the target variable. Together, these improvements confirm that hyperparameter tuning enhanced model quality.

6. Final Model Justification

The tuned Decision Tree Regressor was selected as the final model because it demonstrated superior performance compared with the baseline model. The model achieved lower prediction error and higher explanatory power while also reducing the risk of overfitting. Several factors support this selection: 1. Improved RMSE indicates more accurate house price predictions. 2. Higher R² score reflects better learning of underlying data relationships. 3. Hyperparameters were selected using GridSearchCV rather than manual trial-and-error. 4. Cross-validation was incorporated into the evaluation process, increasing confidence in model reliability. 5. The tuned model balances predictive performance and generalization more effectively than the baseline model. As a result, the tuned Decision Tree Regressor is considered the most appropriate model for deployment and future prediction tasks within this project.

7. Conclusion

This task demonstrated the importance of model validation, overfitting detection, and hyperparameter tuning in machine learning workflows. Initial evaluation revealed overfitting in the baseline Decision Tree model. Through cross-validation and GridSearchCV, a better-performing model configuration was identified. The final tuned model achieved an RMSE of 0.6389 and an R² score of 0.6885, outperforming the baseline model and providing a more reliable solution for house price prediction.